percent level in year two. These simplifying assumptions will illustrate the point without a loss of generality. For the total-returns portfolio, the portfolio distribution in year two after the 2 percent COLA is $40,800. This represents 4.08 percent from a $1 million portfolio. Meanwhile, for the time segmentation strategy, after building the bond ladder, the growth portfolio has $608,000 left. The distribution from this portfolio is the amount needed to support spending in year eleven. This is $40,000 with eleven years of 2 percent COLAs, then discounted for ten years by the 2.45 percent discount rate. This is $39,043, which reflects a much higher 6.42 percent distribution rate from the remaining growth portfolio. The automatic rolling ladder approach does not provide discretion to avoid extending the bond ladder, and for the example illustrated here, it ends up forcing too much distribution pressure on the remaining growth portfolio to be an effective strategy. It actually enhances sequence-of-returns risk.

The total-returns approach with matching dynamic asset allocation performs a bit better since it does not force the distribution rate as high so quickly, but this asset allocation cannot support spending as well as a fixed 60.8 percent stock allocation. When the portfolio is endangered, the strategy pushes the stock allocation down, which is effectively locking failure into place for a greater number of the simulated market scenarios.

Exhibit 7.21

Probability of Success for Different Investment Strategies in Retirement

To Support an Initial $40,000 Spending Goal with a 2% COLA